PROPERTIES AND UNIQUENESS OF THE INJECTIVE HULL OF A MODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that the injective hull of a module is an essential extension and that essential extensions are characterised by non-annihilated multiples. Invoked inside the counted proof (source lines 743-745 and 750-751). Gives essentialness of the hull and of the larger module.

Source reference. Original lemma lemma-essential: sources/stacks-dualizing/pinned-original-source.tex, source lines 338–338. Navigation pointer only: the counted text refers to the Stacks item lemma-essential, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let R be a ring. Let M, N be R-modules and let $M \to E$ and $N \to E'$ be injective hulls. Then*

(1) *for any R-module map $\varphi : M \to N$ there exists an R-module map $\psi : E \to E'$ such that*

$$\begin{array}{ccc} M & \longrightarrow & E \\ {\scriptstyle\varphi}\downarrow & & \downarrow{\scriptstyle\psi} \\ N & \longrightarrow & E' \end{array}$$

commutes,

(2) *if φ is injective, then ψ is injective,*

(3) *if φ is an essential injection, then ψ is an isomorphism,*

(4) *if φ is an isomorphism, then ψ is an isomorphism,*

(5) *if $M \to I$ is an embedding of M into an injective R-module, then there is an isomorphism $I \cong E \oplus I'$ compatible with the embeddings of M,*

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In particular, the injective hull E of M is unique up to isomorphism.

Proof. Part (1) follows from the fact that E' is an injective R-module. Part (2) follows as $\mathrm{Ker}(\psi) \cap M = 0$ and E is an essential extension of M. Assume φ is an essential injection. Then $E \cong \psi(E) \subset E'$ by (2) which implies $E' = \psi(E) \oplus E''$ because E is injective. Since E' is an essential extension of M (Lemma lemma-essential) we get $E'' = 0$. Part (4) is a special case of (3). Assume $M \to I$ as in (5). Choose a map $\alpha : E \to I$ extending the map $M \to I$. Arguing as before we see that α is injective. Thus as before $\alpha(E)$ splits off from I. This proves (5). $\square$